CS 57800 – Statistical Machine Learning

Lecture 3 Probabilistic Inference and Statistical Physics

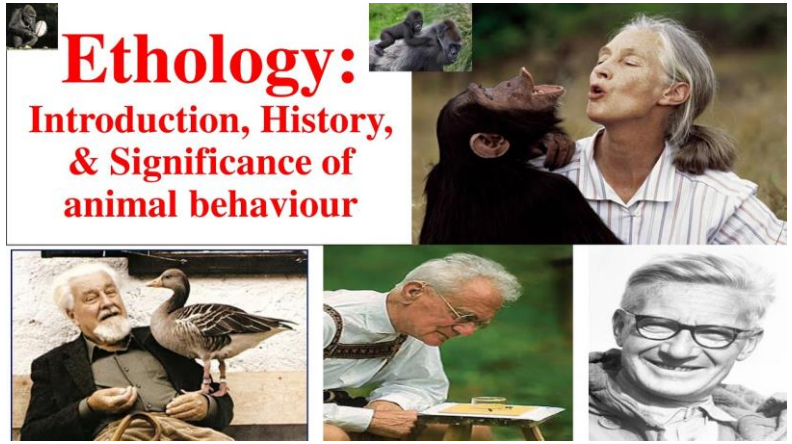
https://ruizhezhang.com/course_fall_2026.html

“Physics” of ML/AI

Math

- Fully rigorous theorems!
- Too idealistic assumptions
- Too simple models (e.g., 2-layer transformer)
- Hard to connect to practice
- Progress very slow

“Ethology”



“Physics” of ML/AI

Math

- Fully rigorous theorems!
- Too idealistic assumptions
- Too simple models (e.g., 1-layer transformer)
- Hard to connect to practice
- Progress very slow

“Ethology”

- Can study large models (GPT 6, Llama 4, …)
- Can be very educational (CoT, ToT, MoE, …)
- Less rigorous, too subjective
- Sometimes lead to false claims

“Physics” of ML/AI

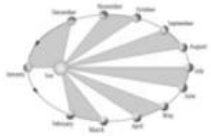
Math

“Ethology”

Newton's laws



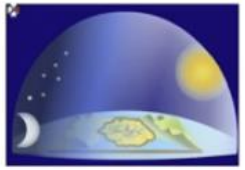
Kepler's laws of planetary motion



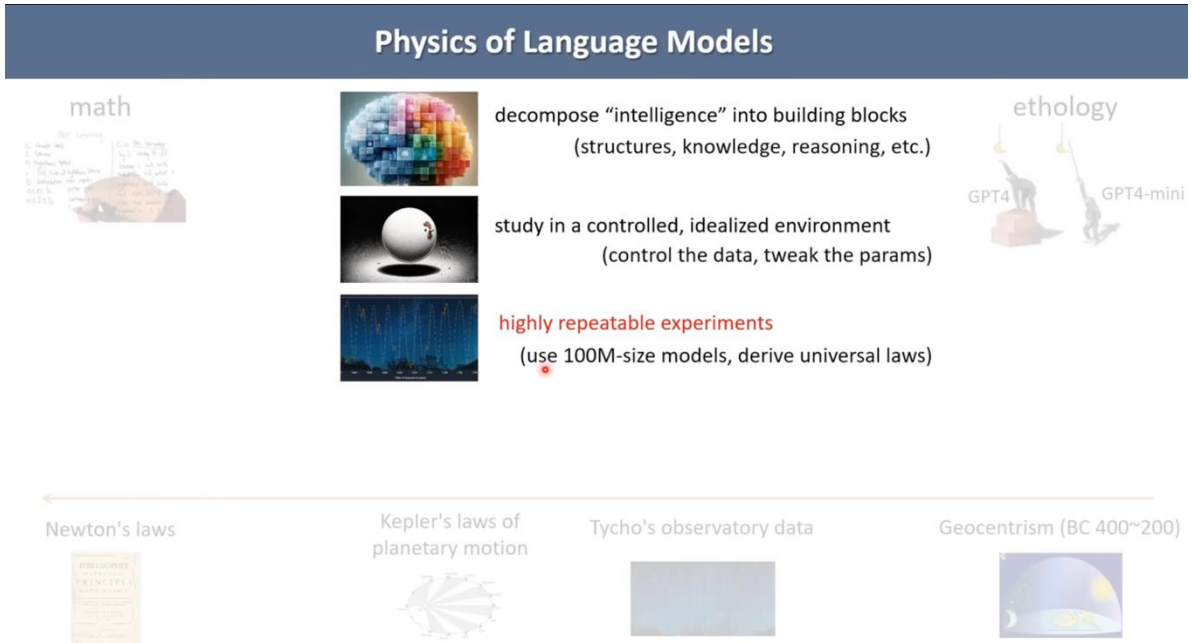
Tycho's observatory data



Geocentrism (BC 400~200)



“Physics” of ML/AI



Physics of Language Models Zeyuan Allen-Zhu

(https://physics.allen-zhu.com/)



ICML 2024
International Conference On Machine Learning

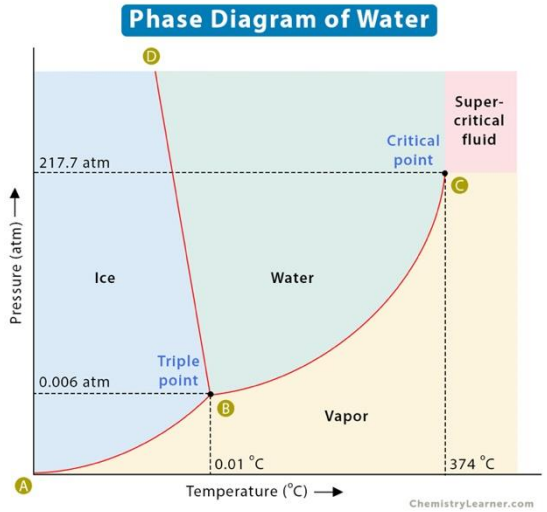
“Physics” of ML/AI

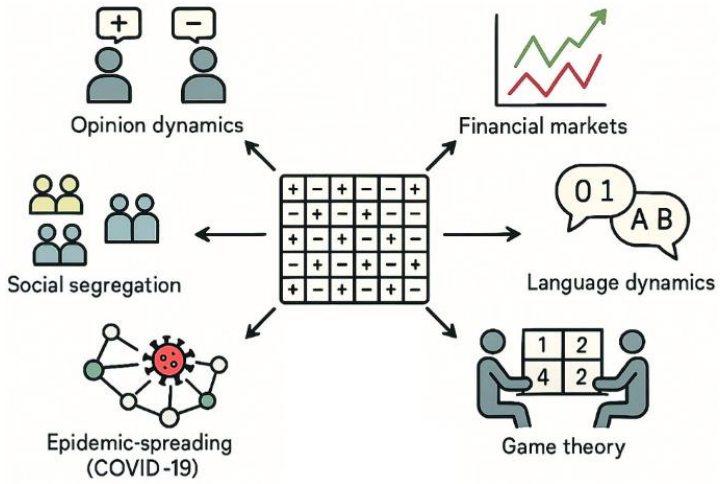
- Our world is super complicated
- Theoretical physicists are very good at working on toy models to understand the world

Example: the Ising model

$$H_N(\boldsymbol{S}) = -J\sum_{\{i,j\}} S_i S_j - h\sum_i S_i\,, \qquad \boldsymbol{S} \in \{-1,1\}^N$$

- Can demonstrate phase transitions (e.g., water freezing into ice, a metal losing its magnetism when heated, etc.)



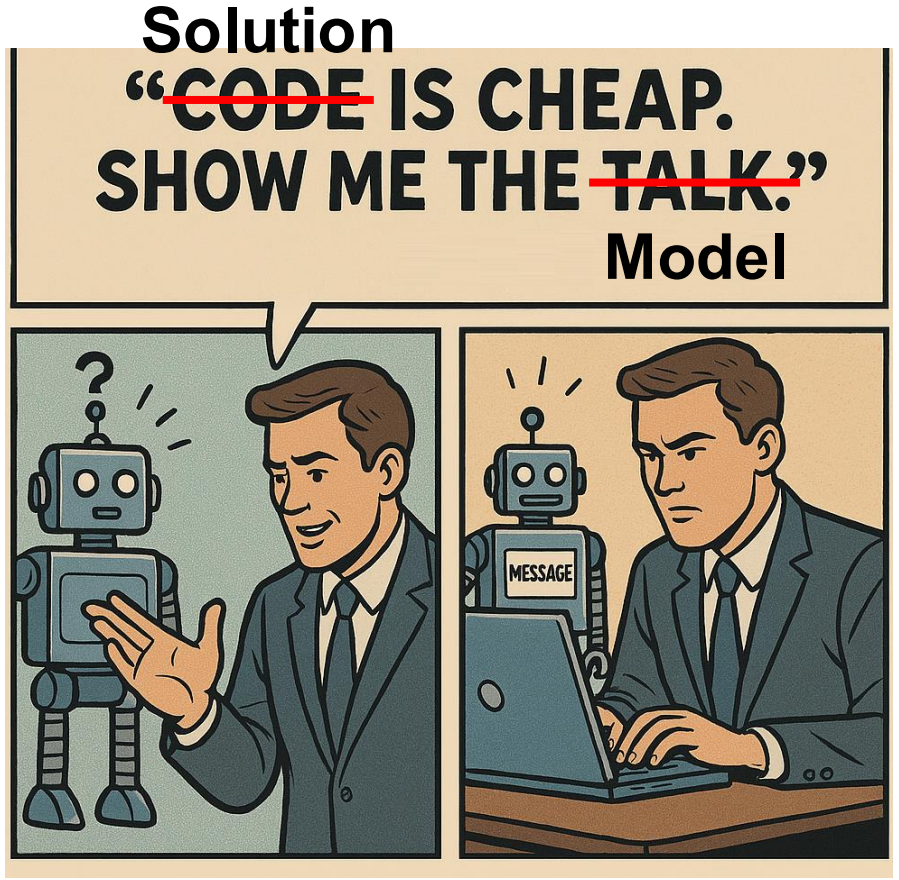


“Physics” of ML/AI

- Our world is super complicated
- Theoretical physicists are very good at working on toy models to understand the world

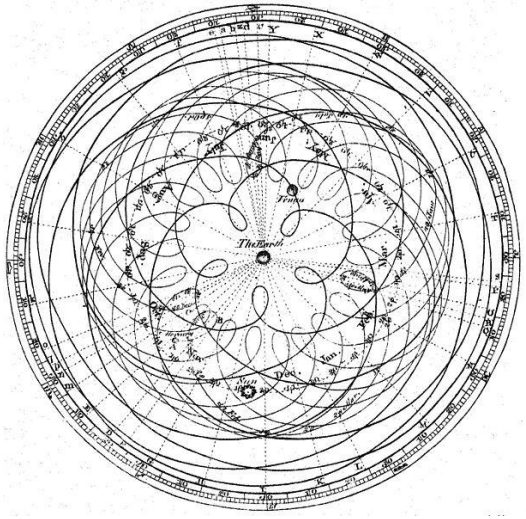
Personal thought:

- In the AGI-era,
- solving a model is much cheaper,
- identifying a useful model (≈problem) is more valuable



ML for Physics

- Physicists have done “machine learning” (inference) over centuries



The Ptolemaic epicycle model

(Overfitting)

Linear regression,
Newton’s method,
etc.

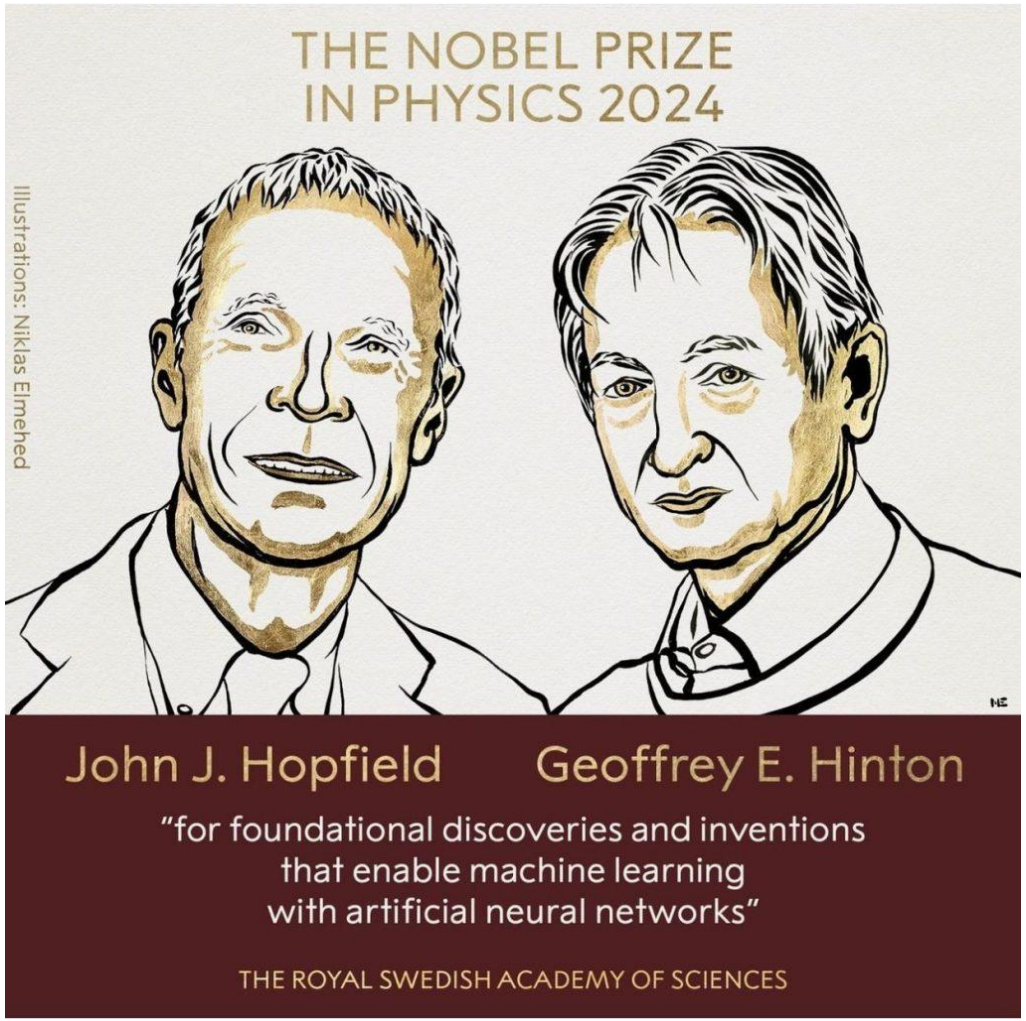
New data

$$F = G\frac{m_1 m_2}{r^2}$$

Newton's law of universal gravitation

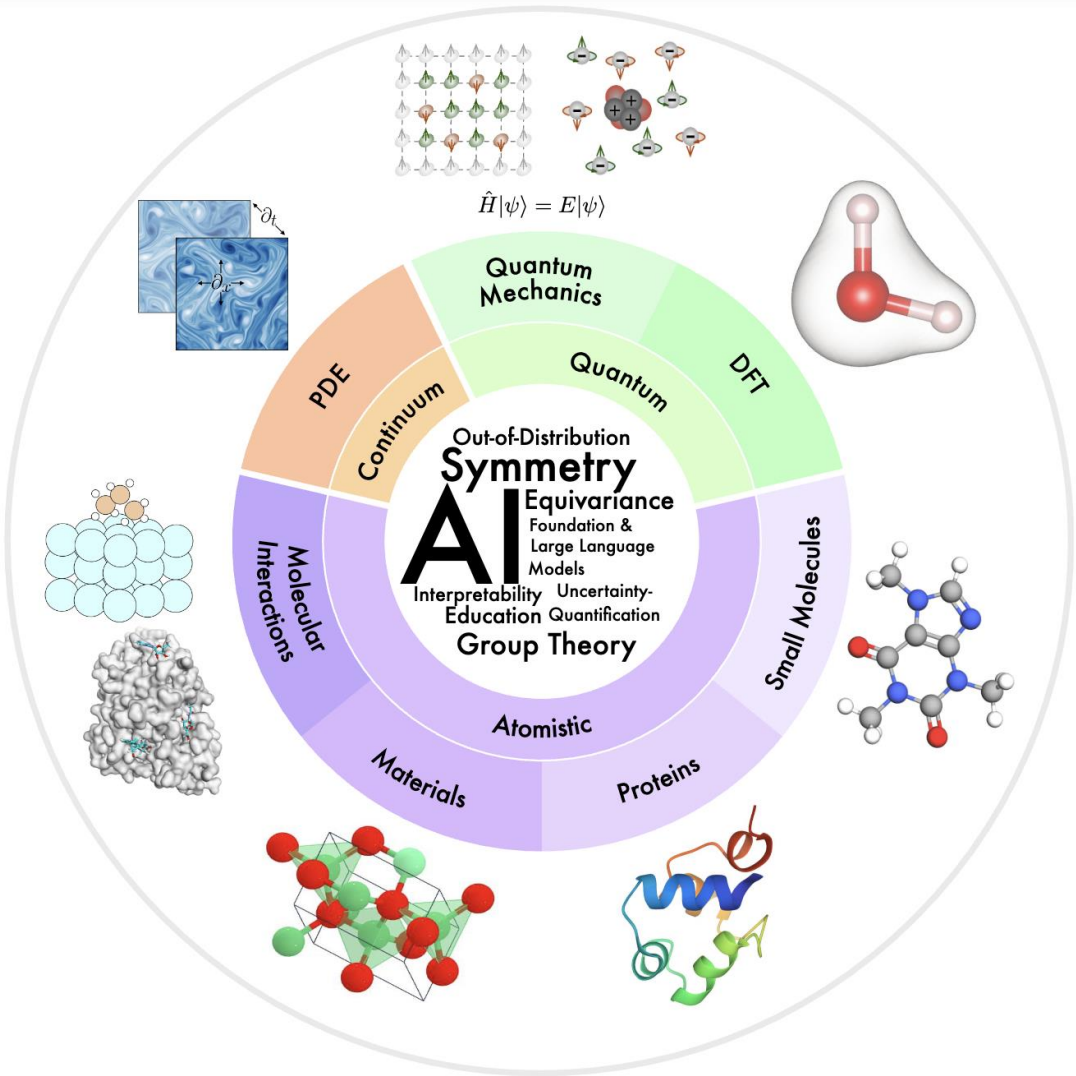
ML for Physics

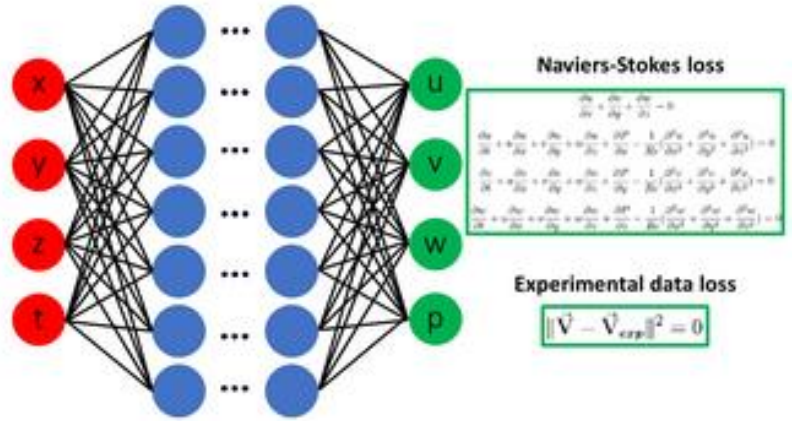
Hopfield network

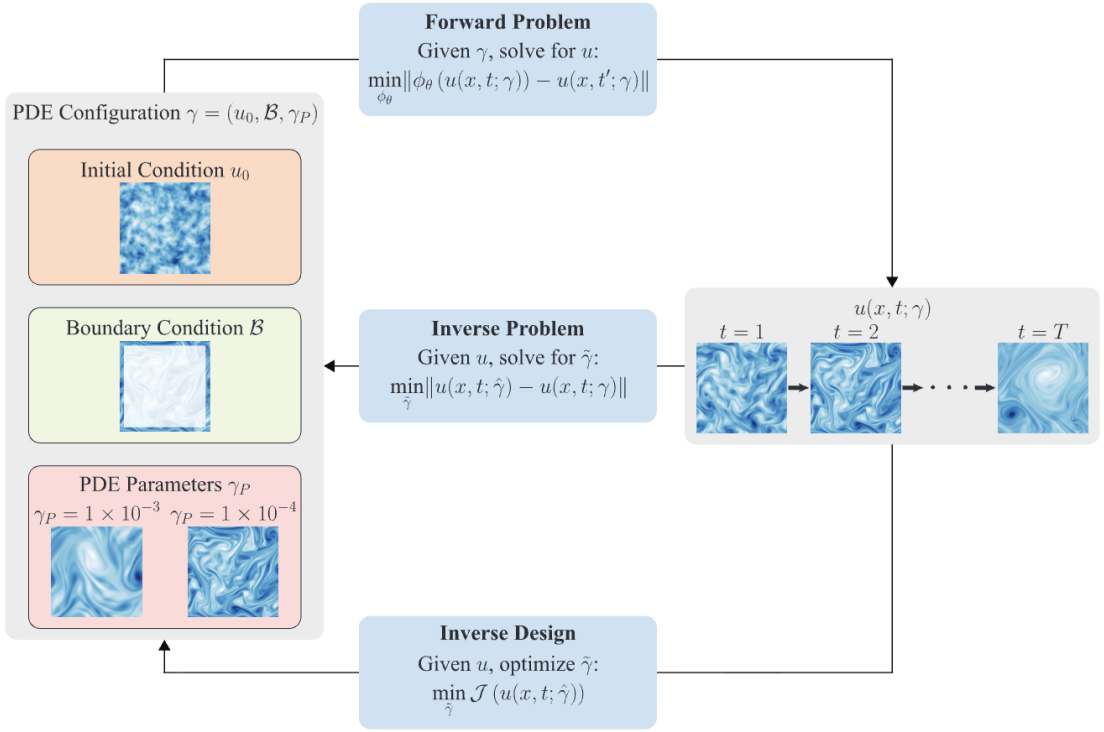


Boltzmann machine

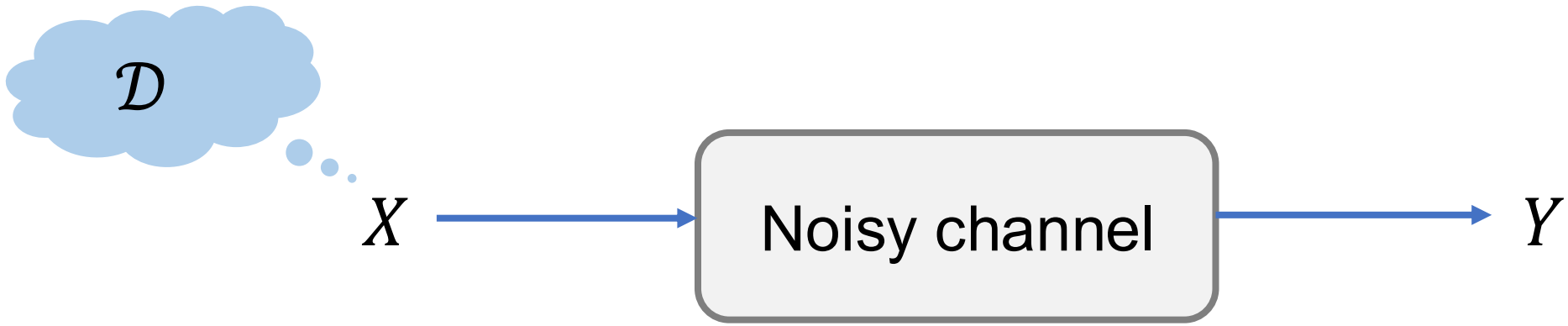
ML for Physics







Physics & Inference



Observing Y, how can you infer X?

- **Teacher-student model for neural network learning:** X is the weight of a neural network F_X. Giving data $\{(x_i, y_i) : x_i \sim \mathcal{N}(0, I_d), y_i = F_X(x_i)\}$, can you learn the neural network weight X?
- **Image denoising:** X is an image from a distribution $\mathcal{D}$, Y is the image with added noise to pixels. Can you recover the original image?

Physics & Inference

The most widely used inference method is the Bayes rule:

$$\Pr[X = x|Y = y] = \frac{\Pr[Y = y|X = x]\Pr[X = x]}{\Pr[Y = y]}$$

- $\Pr[X = x]$: our prior on X
- $\Pr[Y = y|X = x]$: likelihood of the noisy channel
- $\Pr[Y = y]$: normalization constant (or called partition function in physics)

$$Z \coloneqq \Pr[Y = y] = \int \Pr[Y = y|X = x]\Pr[X = x]\,\mathrm{d}x$$

- **Boltzmann energy distribution:**

$$\Pr[X = x|Y = y] = \frac{1}{Z}e^{-\mathcal{E}(x)}$$

$$\mathcal{E}(x) \coloneqq -\log(\Pr[Y = y|X = x]\Pr[X = x])$$

References

- Marc Mezard and Andrea Montanari. *Information, physics, and computation.* 2009.
- Florent Krzakala and Lenka Zdeborová. *Statistical Physics Methods in Optimization and Machine Learning: An Introduction to Replica, Cavity & Message-Passing techniques.* 2022.
- Martin J. Wainwright and Michael Irwin Jordan. *Graphical models, exponential families, and variational inference.* 2008.
- Andrea Montanari and Subhabrata Sen. *A short tutorial on mean-field spin glass techniques for non-physicists.* 2022.

Undirected Graphical Models

- Let $\{\psi_{ij}\}_{(ij)\in F}$ be the compatibility function $\{-1,1\}^2 \to \mathbb{R}_{\geq 0}$ that dictate interactions between particles
- F is a family of edges
- The Gibbs measure μ is a distribution over $\{-1,1\}^n$ given by

$$\mathcal{E}(\boldsymbol{x}) = -\sum_{(ij)\in F} \log\left(\psi_{ij}\left(x_i, x_j\right)\right),$$

$$\mu(\boldsymbol{x}) = \frac{1}{Z} e^{-\mathcal{E}(\boldsymbol{x})} = \frac{1}{Z} \prod_{(ij)\in F} \psi_{ij}\left(x_i, x_j\right),$$

$$Z = \sum_{\boldsymbol{x}\in\{-1,1\}^n} e^{-\mathcal{E}(\boldsymbol{x})}.$$

- $\boldsymbol{x} \in \{-1,1\}^n$ is a configuration of spins

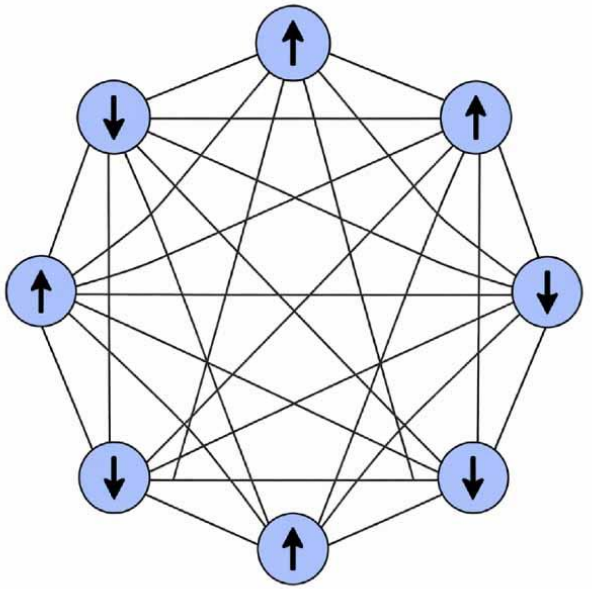
Undirected Graphical Models

Ising models

- $\psi_{ij}(x_i, x_j) = \exp(\beta x_i A_{ij} x_j)$ for some matrix A such that $A_{ii} = 0 \; \forall i \in [n]$ (why?) and A is symmetric

$$\mu(\boldsymbol{x}) \propto \exp\left(+\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x}\right)$$

- β is the inverse temperature
- A is the Hamiltonian or the interaction matrix of the system
- $\mathcal{E}(\boldsymbol{x}) = -\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x}$ is the energy of the system
- When $\beta \to 0$ (high-temperature limit), $\mu \to \mathrm{Unif}(\{-1,1\}^n)$
- When $\beta \to \infty$ (low-temperature limit), $\mu \to \mathrm{Unif}(\arg\min \mathcal{E}(\boldsymbol{x}))$

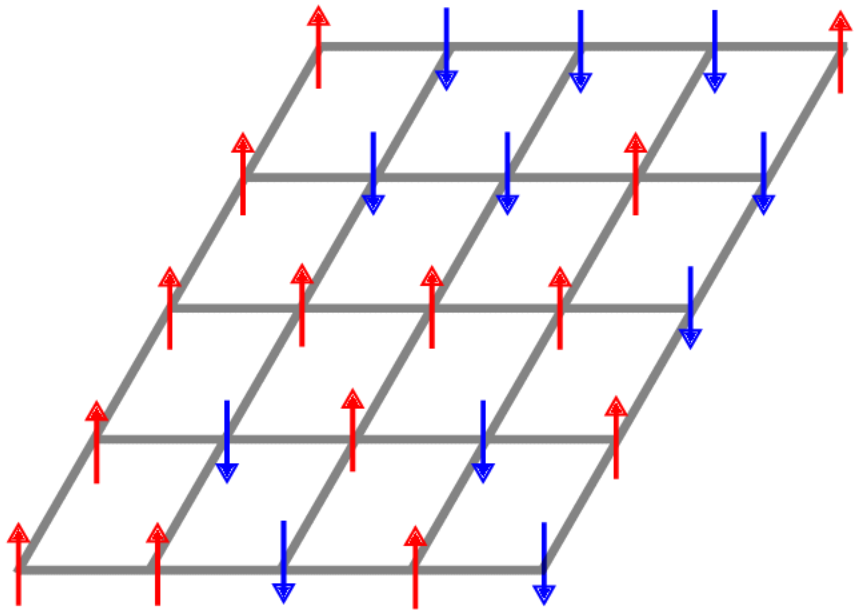


A defines a weighted graph G

Undirected Graphical Models

$$\mu(\boldsymbol{x}) = \frac{\exp\left(\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x}\right)}{Z}, \qquad Z = \sum_{\boldsymbol{x}\in\{-1,1\}^n} \exp\left(\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x}\right)$$

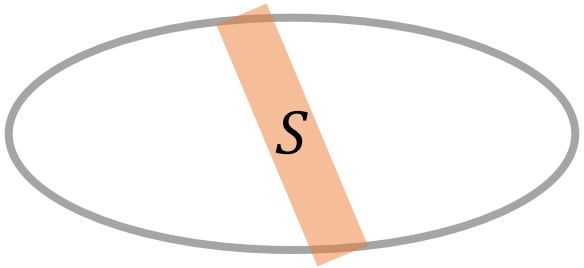
- Suppose A is the adjacent matrix of a 2D lattice (unit weight)
- $\mathcal{E}(\boldsymbol{x}) = -\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x} = -\frac{\beta}{2}\sum_{(i,j)\in F} x_i x_j$
- $\mathcal{E}(\boldsymbol{x})$ is minimized when $x_1 = x_2 = \cdots = x_n = \pm 1$



Undirected Graphical Models

Markov property

- For $S \subseteq [n]$ such that $[n] \backslash S$ decomposes into disjoint pieces in G, then conditioned on any assignment to the spins in S, the marginal distributions on the pieces are independent



- For Ising models, for any $i \in [n]$, let $S = \partial_i$ (the neighbors of i in G)

$$\Pr\left[x_i = \sigma \middle| \boldsymbol{x}_{\partial_i} = \boldsymbol{s}\right] \propto \exp\left(-\beta\sigma \sum_{j \in \partial_i} A_{ij} s_j\right)$$

Undirected Graphical Models

Markov property

- For $S \subseteq [n]$ such that $[n] \backslash S$ decomposes into disjoint pieces in G, then conditioned on any assignment to the spins in S, the marginal distributions on the pieces are independent

- Global Markov property: the above condition holds for any $S \subseteq [n]$
- Local Markov property: the above condition holds for $S = \partial_i \; \forall i \in [n]$
- Pairwise Markov property: the above condition holds for $S = [n] \backslash \{i, j\}$ for any non-adjacent $i, j \in [n]$

Global Markov $\Longrightarrow$ Local Markov $\Longrightarrow$ Pairwise Markov

Hammersley–Clifford theorem: if $\mu(\boldsymbol{x}) > 0$ for any $\boldsymbol{x} \in \{-1,1\}^n$ (i.e., a Gibbs distribution),
then Pairwise Markov $\Longrightarrow$ Global Markov

Fundamental Algorithmic Problems in Inference

1. Draw samples from the Gibbs distribution μ
2. Compute the partition function Z

Jerrum, Valiant, and Vazirani '86: the equivalence of sampling and counting

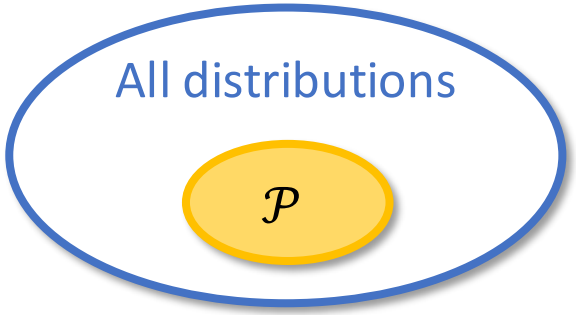
- Computing the partition function exactly is very hard: for $\psi_{ij}(x_i, x_j) = \mathbf{1}_{(x_i,x_j)\neq(1,1)}$, computing Z is counting the number of independent sets of the interaction graph, which is a **#P**-complete problem
- Our goal is to estimate Z approximately. Some approaches:
  - Markov chain Monte Carlo (MCMC)
  - Variational inference (VI)
  - Diffusion models

Variational Inference

Approximate μ by distributions from some family $\mathcal{P}$ of "simpler" distributions that are easy to sample from (e.g., product distributions a.k.a. naïve mean-field):

$$\min_{\nu \in \mathcal{P}} \mathrm{KL}(\nu \| \mu) \geq 0$$

- If $\mathcal{P} = \{\text{all distributions}\}$, then the minimizer is $\nu = \mu$ (Gibbs' inequality)



- **Issue:** the KL optimizer is hard to evaluate

$$\mathrm{KL}(\nu \| \mu) = \mathbb{E}_{x \sim \nu}\left[\log \frac{\nu(x)}{\mu(x)}\right] = \mathbb{E}_{x \sim \nu}\left[\log \frac{\nu(x)}{e^{-\mathcal{E}(x)}/Z}\right] = \underbrace{\mathbb{E}_{x \sim \nu}[\log \nu(x)]}_{-H(\nu)} + \underbrace{\mathbb{E}_{x \sim \nu}[\mathcal{E}(x)]}_{\text{Averaged energy}} + \log Z$$

- Define the Gibbs free energy $G[\nu] \coloneqq -H(\nu) + \mathbb{E}_{x \sim \nu}[\mathcal{E}(x)]$ $-1\times$ "evidence lower bound" (ELBO)
- To minimize the KL divergence, it suffices to minimize the free energy $G[\nu]$

Detour: Evidence Lower Bound

$$G[q] \coloneqq -H(q) + \mathbb{E}_{\boldsymbol{x}\sim q}[\mathcal{E}(\boldsymbol{x})], \qquad \mathcal{E}(\boldsymbol{x}) = -\log \Pr\big[[X = \boldsymbol{x}, Y = \boldsymbol{y}]\big]$$

$$\mathrm{KL}\big(q(X)\big\|\Pr[X|Y = \boldsymbol{y}]\big) = \mathbb{E}_{\boldsymbol{x}\sim q}[\log q(\boldsymbol{x})] - \mathbb{E}_{\boldsymbol{x}\sim q}[\log \Pr[X = \boldsymbol{x}, Y = \boldsymbol{y}]] + \log \Pr[Y = \boldsymbol{y}]$$

$$\log \Pr[Y = \boldsymbol{y}] = \mathbb{E}_{\boldsymbol{x}\sim q}[\log \Pr[X = \boldsymbol{x}, Y = \boldsymbol{y}]] - \mathbb{E}_{\boldsymbol{x}\sim q}[\log q(\boldsymbol{x})] + \mathrm{KL}\big(q(X)\big\|\Pr[X|Y = y]\big)$$

$$\mathrm{ELBO}(q) \coloneqq \mathbb{E}_{\boldsymbol{x}\sim q}[\log \Pr[X = \boldsymbol{x}, Y = \boldsymbol{y}]] - \mathbb{E}_{\boldsymbol{x}\sim q}[\log q(\boldsymbol{x})] = \mathbb{E}_{\boldsymbol{x}\sim q}\left[\frac{\log \Pr[X = \boldsymbol{x}, Y = \boldsymbol{y}]}{\log q(\boldsymbol{x})}\right]$$

- $\mathrm{ELBO}(q)$ is a lower bound for the log-evidence $\log \Pr[Y]$
- Frequently used in ML practice (e.g., variational autoencoder, diffusion models, EM algorithms, …)

Variational Inference

$$G[\nu] \coloneqq -H(\nu) + \mathbb{E}_{\boldsymbol{x}\sim\nu}[\mathcal{E}(\boldsymbol{x})]$$

Consider the Ising model:

$$\mu(\boldsymbol{x}) \propto \exp\left(\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x}\right)$$

$$G[\nu] = -\mathbb{E}_{\boldsymbol{x}\sim\nu}\left[\frac{\beta}{2}\boldsymbol{x}^\top A\boldsymbol{x}\right] - H(\nu)$$

Entropy regularization

- When β is large, minimization prioritizes minimizing the averaged energy
- When β is small, minimization prioritizes maximizing the entropy
- For a fixed $\nu \in \mathcal{P}$, $G[\nu]$ is easy to evaluate. But it is hard to optimize a priori
- We will introduce a general approach: the belief propagation (BP), to minimize $G[\nu]$

Belief Propagation

- Belief propagation (BP) is a powerful, general heuristic to solve $\min_{\nu} G[\nu]$
- When the interaction graph is a tree, BP is exact

Two interpretations of BP:

1. Dynamic programming on a tree
2. Nonconvex optimization for finding a stationary point of a relaxation of the Gibbs free energy (Bethe free energy)

In the following discussions, we choose a canonical example: random independent sets

- Let $G = (V, E)$ be a graph (in particular, a tree)
- The compatibility function is $\psi_{ij}(x_i x_j) = \mathbf{1}_{(x_i, x_j) \neq (1,1)}$ for every edge $(i, j) \in E$

Belief Propagation

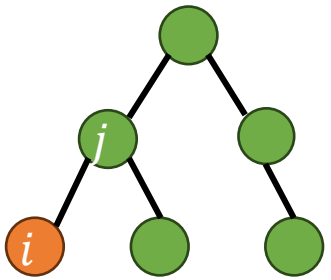
- The Gibbs free energy on a tree only depends on 1- and 2-wise marginals of μ

Lemma. If the interaction graph $G = (V, E)$ is a tree, then for any distribution μ on G,

$$\mu(\boldsymbol{x}) = \prod_{(i,j)\in E} \mu_{ij}(x_i, x_j) \prod_{i\in V} \mu_i(x_i)^{1-|\partial i|}$$

Proof.

- We prove by induction: $|V| = 1$ is trivially true
- For $|V| > 1$, let i be any leaf node of the tree connected by (i, j)



$$\begin{aligned}
\Pr_{\boldsymbol{x}\sim\mu}[\boldsymbol{x} = s] &= \Pr_{\mu}\left[\boldsymbol{x}_{V\setminus\{i\}} = \boldsymbol{s}_{V\setminus\{i\}}\right] \Pr_{\mu}\left[x_i = s_i \middle| \boldsymbol{x}_{V\setminus\{i\}} = \boldsymbol{s}_{V\setminus\{i\}}\right] \\
&= \Pr_{\mu}\left[\boldsymbol{x}_{V\setminus\{i\}} = \boldsymbol{s}_{V\setminus\{i\}}\right] \Pr_{\mu}\left[x_i = s_i \middle| x_j = s_j\right] \qquad \text{(Markov property)} \\
&= \Pr_{\mu}\left[\boldsymbol{x}_{V\setminus\{i\}} = \boldsymbol{s}_{V\setminus\{i\}}\right] \mu_{ij}(s_i, s_j)/\mu_j(s_j)
\end{aligned}$$

Belief Propagation

Proof.

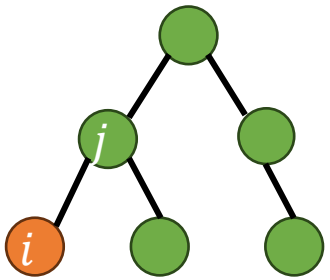
- By the induction hypothesis,

$$\Pr_{\mu}\left[\boldsymbol{x}_{[n]\setminus\{i\}} = \boldsymbol{s}_{[n]\setminus\{i\}}\right] = \prod_{(k,\ell)\in E\setminus(i,j)} \mu_{k\ell}(s_k, s_\ell) \prod_{k\in V\setminus\{i\}} \mu_k(s_k)^{1-|\partial_{V\setminus\{i\}}k|}$$

- Thus,

$$\Pr_{\boldsymbol{x}\sim\mu}[\boldsymbol{x} = s] = \prod_{(k,\ell)\in E\setminus(i,j)} \mu_{k\ell}(s_k, s_\ell) \prod_{k\in V\setminus\{i\}} \mu_k(s_k)^{1-|\partial_{V\setminus\{i\}}k|} \cdot \frac{\mu_{ij}(s_i, s_j)}{\mu_j(s_j)}$$

$$= \prod_{(k,\ell)\in E} \mu_{k\ell}(s_k, s_\ell) \prod_{k\in V} \mu_k(s_k)^{1-|\partial_V k|}$$



■

Belief Propagation

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$$\mu(\boldsymbol{x}) = \prod_{(i,j)\in E} \mu_{ij}(x_i, x_j) \prod_{i\in V} \mu_i(x_i)^{1-|\partial i|}$$

- The Gibbs free energy is defined as

$$G[\mu] \coloneqq -H(\mu) + \mathbb{E}_{\boldsymbol{x}\sim\mu}[\mathcal{E}(\boldsymbol{x})]$$

$$\mathbb{E}_{\boldsymbol{x}\sim\mu}[\mathcal{E}(\boldsymbol{x})] = \sum_{(i,j)\in E} -\mathbb{E}_{\mu_{ij}}[\log \psi_{ij}(x_i, x_j)]$$

$$-H(\mu) = \mathbb{E}_{\boldsymbol{x}\sim\mu}[\log \mu(\boldsymbol{x})] = \sum_{(i,j)\in E} \mathbb{E}_{\mu_{ij}}[\log \mu_{ij}(x_i, x_j)] + \sum_{i\in V} (1 - |\partial i|) \cdot \mathbb{E}_{\mu_i}[\log \mu_i(x_i)]$$

Belief Propagation

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$$\mu(\boldsymbol{x}) = \prod_{(i,j)\in E} \mu_{ij}(x_i, x_j) \prod_{i\in V} \mu_i(x_i)^{1-|\partial i|}$$

- Thus, we have

$$G[\mu] = \sum_{(i,j)\in E} \mathbb{E}_{\mu_{ij}}\left[\frac{\log \mu_{ij}(x_i, x_j)}{\log \psi_{ij}(x_i, x_j)}\right] + \sum_{i\in V}(1 - |\partial i|) \cdot \mathbb{E}_{\mu_i}[\log \mu_i(x_i)]$$

- It suffices to compute the marginals μ_i and μ_{ij}

Remark: The marginal expectation $\{\mathbb{E}_{\mu_i}[x_i]\}_{i\in[n]}$ is also the MMSE estimator:

$$\arg\min_{\hat{x}(\cdot)} \mathbb{E}_{X,Y}[\|X - \hat{x}(Y)\|^2] = \mathbb{E}_Y\left[\mathbb{E}_{X|Y}[\|X - \hat{x}(Y)\|^2]\right]$$

Belief Propagation

Random Independent Sets

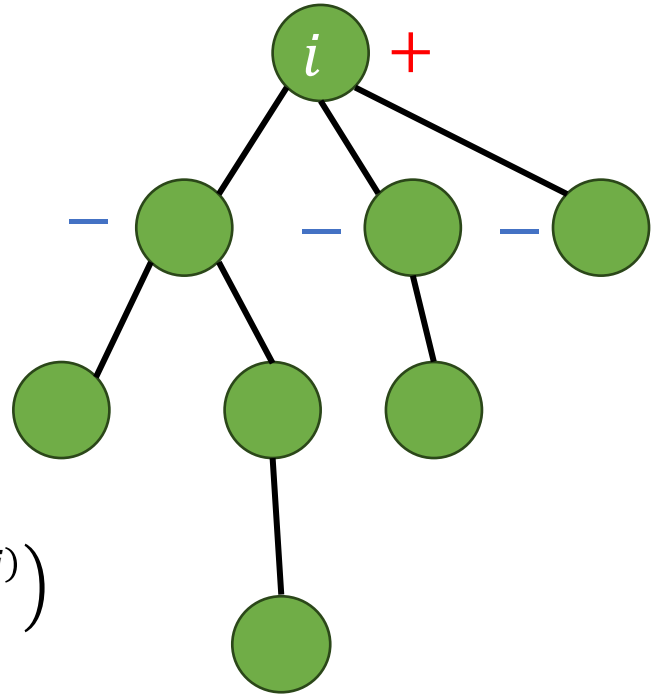
- Let $G = (V, E)$ be a graph (in particular, a tree)
- The compatibility function is $\psi_{ij}(x_i x_j) = \mathbf{1}_{(x_i, x_j) \neq (1,1)}$ for every edge $(i, j) \in E$

How to compute $\mu_i(x_i)$ on a tree?

Dynamic programming!

- We can take i to be the root
- Let $Z_\sigma^{(i)} :=$ the number of independent sets in the (sub-)tree with root i being labelled $\sigma \in \{-1, 1\}$
- We have the DP recursion:

$$Z_+^{(i)} = \prod_{j \in \partial i} Z_-^{(j)}, \qquad Z_-^{(i)} = \prod_{j \in \partial i} \left(Z_-^{(j)} + Z_+^{(j)} \right)$$



Belief Propagation

Random Independent Sets

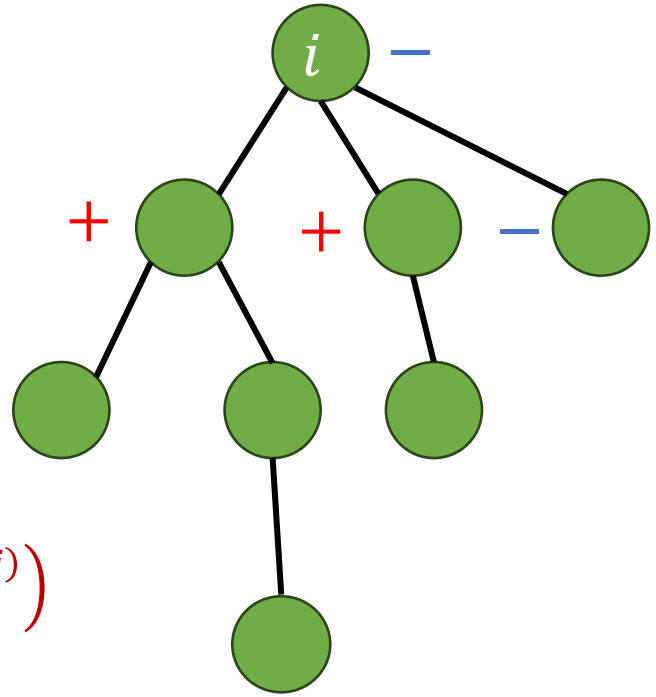
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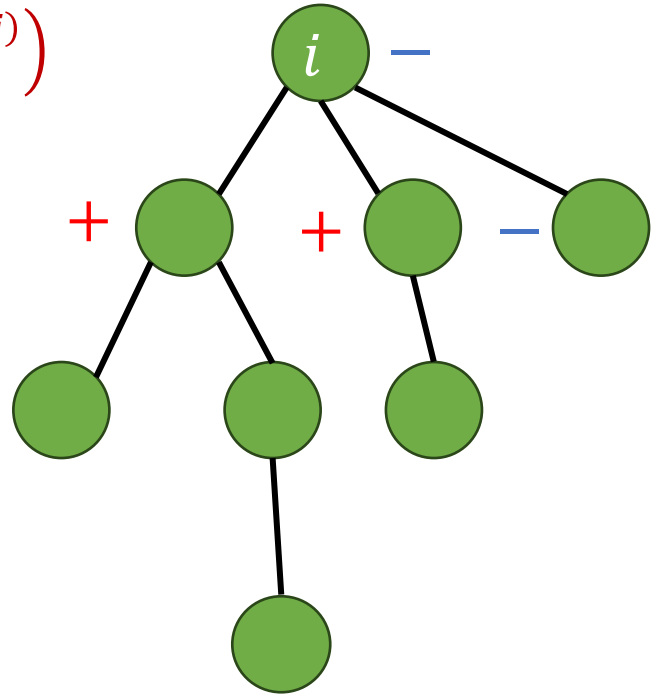
$$Z_+^{(i)} = \prod_{j \in \partial i} Z_-^{(j)}, \qquad Z_-^{(i)} = \prod_{j \in \partial i} \left(Z_-^{(j)} + Z_+^{(j)} \right)$$



Belief Propagation

- We can take i to be the root
- Let $Z_\sigma^{(i)} :=$ the number of independent sets in the (sub-)tree with root i being labelled $\sigma \in \{-1,1\}$
- We have the DP recursion:

$$Z_+^{(i)} = \prod_{j \in \partial i} Z_-^{(j)}, \qquad Z_-^{(i)} = \prod_{j \in \partial i} \left(Z_-^{(j)} + Z_+^{(j)} \right)$$



- The marginal is

$$\Pr[x_i = +1] = \frac{Z_+^{(i)}}{Z_+^{(i)} + Z_-^{(i)}} \propto Z_+^{(i)}$$

$$\Pr[x_i = -1] \propto Z_-^{(i)}$$

- This DP relies on a global top-down tree structure
  BP takes a more local view (which makes it work for general graphs)

Belief Propagation

We need to define several notations

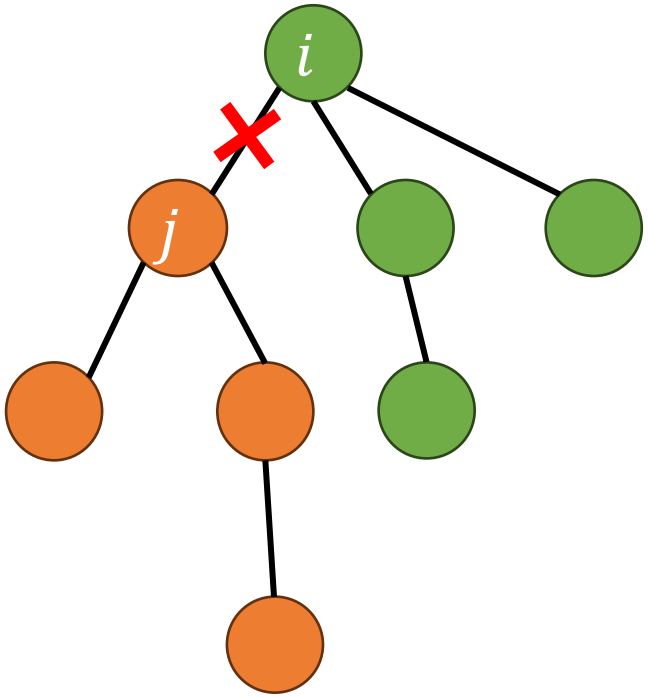
- Let $V^{j\to i}$ denote the sub-tree by cutting the edge (i,j) and keeping the part containing j
- Let $m_\sigma^{j\to i}$ denote the probability that $x_j=\sigma$ in $V^{j\to i}$

a "message" that node j passes to node i

$$m_\sigma^{j\to i} \coloneqq \Pr_{\mu_{Vj\to i}}[x_j=\sigma]$$

$$Z_+^{(i)}=\prod_{j\in\partial i} Z_-^{(j)}=\prod_{j\in\partial i}\Pr_{\mu_{Vj\to i}}[x_j=-]\left(Z_-^{(j)}+Z_+^{(j)}\right)$$

$$\propto \prod_{j\in\partial i}\Pr_{\mu_{Vj\to i}}[x_j=-]=\prod_{j\in\partial i} m_-^{j\to i}$$



$$Z_+^{(i)}=\prod_{j\in\partial i} Z_-^{(j)}, \qquad Z_-^{(i)}=\prod_{j\in\partial i}\left(Z_-^{(j)}+Z_+^{(j)}\right)$$

$$\Pr_{\mu_V}[x_i=\sigma]\propto Z_\sigma^{(i)}$$

Belief Propagation

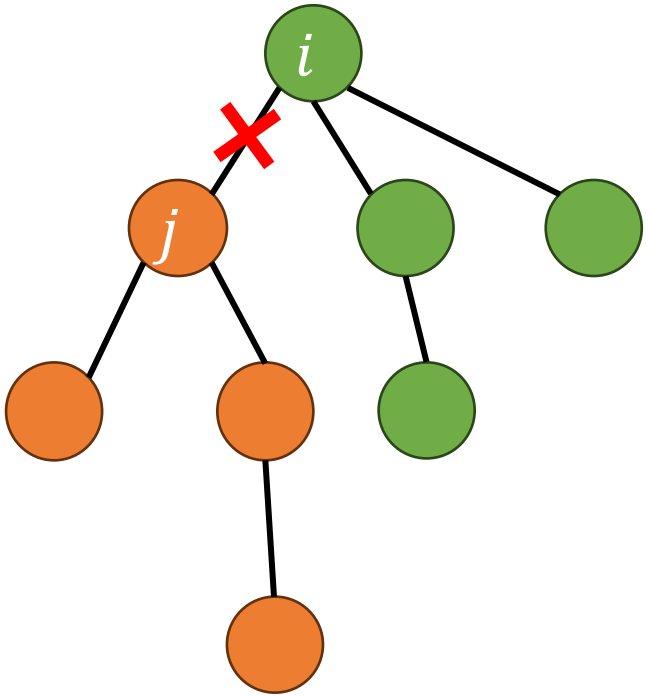
$$\Pr_{\mu}[x_i = \sigma] \propto \sum_{s \in \{\pm 1\}^n : s_i = \sigma} \prod_{(k,\ell) \in E} \psi_{k\ell}(s_k, s_\ell)$$

$$= \sum_{s \in \{\pm 1\}^n : s_i = \sigma} \prod_{j \in \partial i} \left(\psi_{ij}(\sigma, s_j) \prod_{(k,\ell) \in E^{j \to i}} \psi_{k\ell}(s_k, s_\ell) \right)$$

$$= \prod_{j \in \partial i} \sum_{s_{V^{j \to i}}} \left(\psi_{ij}(\sigma, s_j) \prod_{(k,\ell) \in E^{j \to i}} \psi_{k\ell}(s_k, s_\ell) \right)$$

$$= \prod_{j \in \partial i} \sum_{s_j \in \{\pm 1\}} \psi_{ij}(\sigma, s_j) \underbrace{\left(\sum_{s_{V^{j \to i}} \setminus \{j\}} \prod_{(k,\ell) \in E^{j \to i}} \psi_{k\ell}(s_k, s_\ell) \right)}_{\propto m_{s_j}^{j \to i}}$$

$$\boxed{\Pr_{\mu}[x_i = \sigma] \propto \prod_{j \in \partial i} \sum_{s \in \{\pm 1\}} \psi_{ij}(\sigma, s) m_s^{j \to i}}$$



$$\Pr_{\mu}[x_i = \sigma] = \frac{\Pr_{\mu}[x_i = \sigma]}{\Pr_{\mu}[x_i = \sigma] + \Pr_{\mu}[x_i = -\sigma]}$$

Belief Propagation

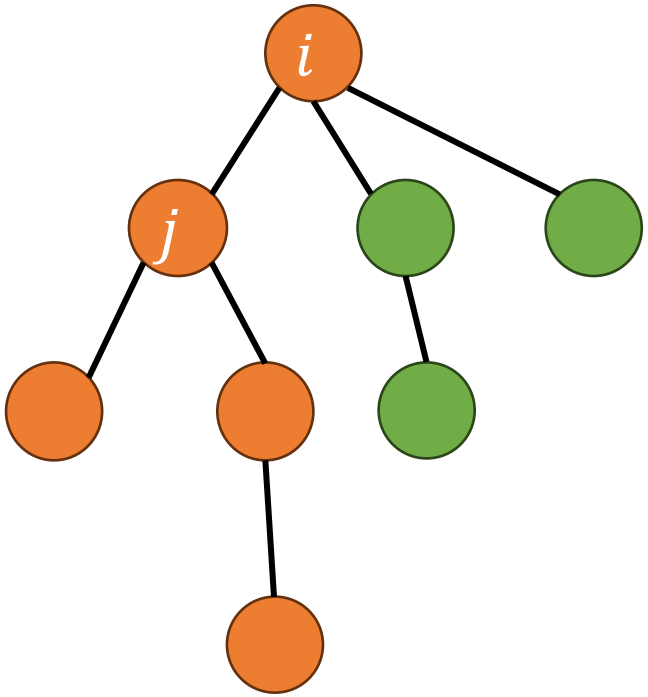
We need to define several notations

- Let $\overline{V}^{j\to i}$ denote the sub-tree by cutting the edge (i,j) and keeping the part containing j and connecting (i,j)
- Let $\overline{m}_\sigma^{j\to i}$ denote the probability that $x_i=\sigma$ in $\overline{V}^{j\to i}$

$$\overline{m}_\sigma^{j\to i} \coloneqq \Pr_{\mu_{\overline{V}^{j\to i}}}[x_i=\sigma]$$

$$\overline{m}_\sigma^{j\to i} \propto \sum_{s\in\{\pm 1\}} \psi_{ij}(\sigma,s) m_s^{j\to i}$$

$$\Pr_{\mu}[x_i=\sigma] \propto \prod_{j\in\partial i} \sum_{s\in\{\pm 1\}} \psi_{ij}(\sigma,s) m_s^{j\to i}$$

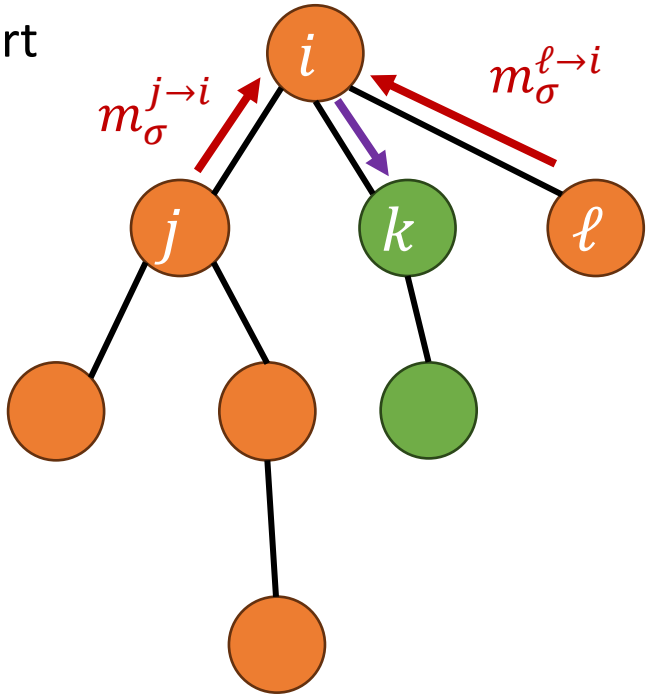


Belief Propagation

We need to define several notations

- Let $\overline{V}^{j\to i}$ denote the sub-tree by cutting the edge (i,j) and keeping the part containing j and connecting (i,j)
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$$\overline{m}_\sigma^{j\to i} \propto \sum_{s\in\{\pm 1\}} \psi_{ij}(\sigma,s) m_s^{j\to i}$$



Given $\left\{m_\sigma^{j\to i} : j\in\partial i, \sigma\in\{\pm 1\}\right\}$, how to compute $m_\sigma^{i\to k}$ for some $k\in\partial i$?

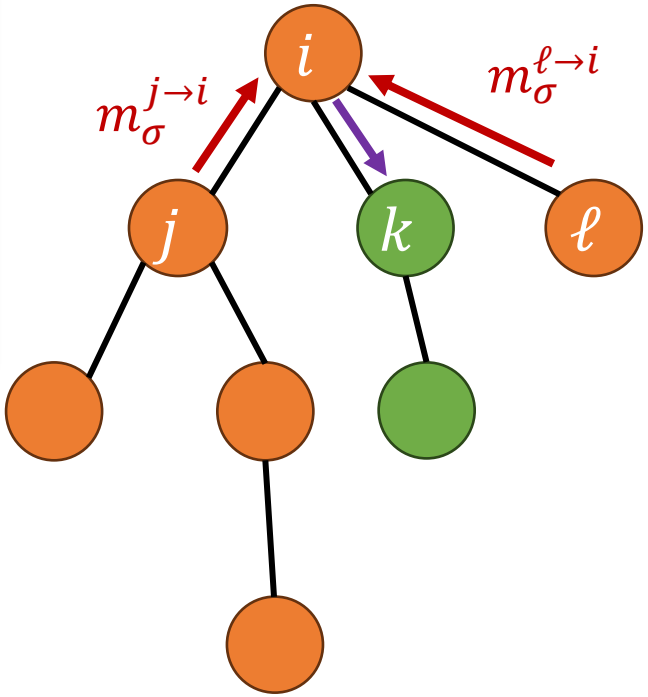
$$m_\sigma^{i\to k} = \Pr_{\mu_{V^{i\to k}}}[x_i=\sigma] \propto \prod_{j\in\partial_{V^{i\to k}} i} \sum_{s\in\{\pm 1\}} \psi_{ij}(\sigma,s) m_s^{j\to i}$$

$$\propto \prod_{j\in\partial i\setminus\{k\}} \overline{m}_\sigma^{j\to i}$$

Belief Propagation

$$m_\sigma^{i\to k} \propto \prod_{j\in\partial i\setminus\{k\}} \overline{m}_\sigma^{j\to i}, \qquad \overline{m}_\sigma^{j\to i} \propto \sum_{s\in\{\pm 1\}} \psi_{ij}(\sigma,s) m_s^{j\to i}$$

$$\Pr_\mu[x_i=\sigma] \propto \prod_{j\in\partial i} \overline{m}_\sigma^{j\to i} \propto m_\sigma^{i\to k}\cdot \overline{m}_\sigma^{k\to i}$$

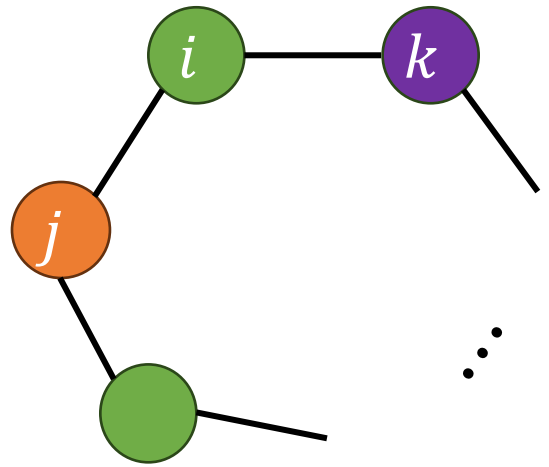


BP algorithm:

- Initialize $m_\sigma^{j\to i} = 1/2$ for all $(i,j) \in E$
- Iteratively update $\left\{m_\sigma^{j\to i}, \overline{m}_\sigma^{j\to i} : (i,j)\in E, \sigma\in\{\pm 1\}\right\}$ in parallel

Example: Belief Propagation on a circle

$$m_{\sigma}^{i\to k} \propto \prod_{j\in\partial i\setminus\{k\}} \sum_{s\in\{\pm 1\}} \psi_{ij}(\sigma,s) m_{s}^{j\to i}$$



- $m_{+}^{i\to k} \propto m_{-}^{j\to i}$
- $m_{-}^{i\to k} \propto m_{+}^{j\to i} + m_{-}^{j\to i}$
- If $\left(m_{+}^{j\to i}, m_{-}^{j\to i}\right) = (p, 1-p)$ before the update, then

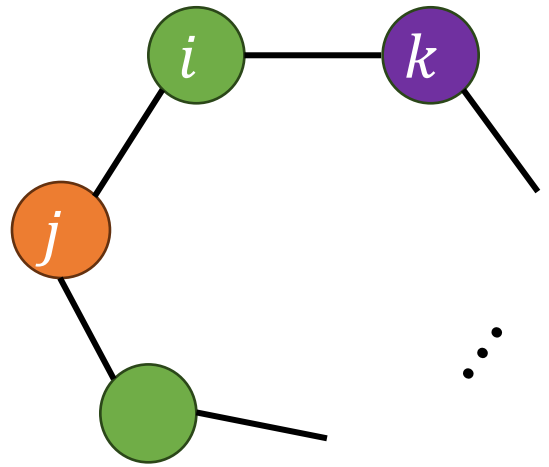
$$m_{+}^{i\to k} = \frac{1-p}{1-p+1} = \frac{1-p}{2-p}, \qquad m_{-}^{i\to k} = \frac{1}{2-p}$$

- We initialize the messages with $p = 1/2$ at the beginning
- After one round of BP updates, $p \longmapsto \frac{1-p}{2-p}$
- $\frac{1}{2} \longrightarrow \frac{1}{3} \longrightarrow \frac{2}{5} \longrightarrow \frac{3}{8} \longrightarrow \frac{5}{13} \longrightarrow \cdots \longrightarrow \frac{1}{\varphi^2} \approx 0.382$

$\varphi = \frac{\sqrt{5}+1}{2}$ is the golden ratio

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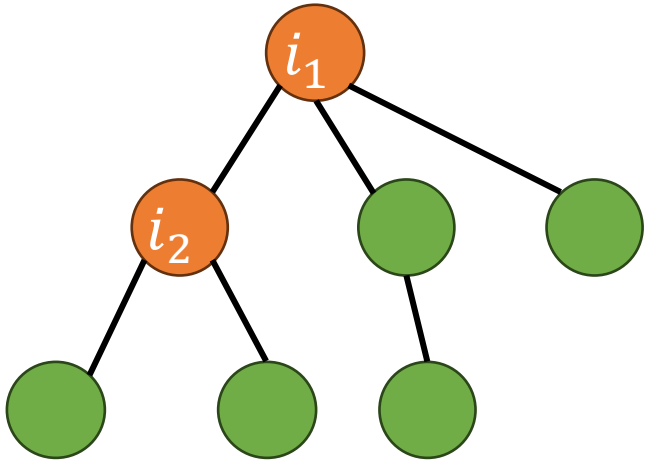
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Belief Propagation

$$\Pr_{\mu}[x_i = \sigma] \propto \prod_{j \in \partial i} \overline{m}_{\sigma}^{j \to i} \propto m_{\sigma}^{i \to k} \cdot \overline{m}_{\sigma}^{k \to i}$$



The messages can also compute 2-wise marginals

- For any $(i_1, i_2) \in E$, let $\partial(i_1, i_2) \coloneqq (\partial i_1 \cup \partial i_2) \backslash \{i_1, i_2\}$
- For $j \in \partial(i_1, i_2)$, let $i(j)$ denote the node such that $\big(j, i(j)\big) \in E$

$$\Pr_{\mu}\big[(x_{i_1}, x_{i_2}) = (\sigma_{i_1}, \sigma_{i_2})\big] \propto \psi_{i_1 i_2}(\sigma_{i_1}, \sigma_{i_2}) \sum_{s_{\partial(i_1,i_2)} \in \{\pm 1\}^{|\partial(i_1,i_2)|}} \prod_{j \in \partial(i_1,i_2)} \Pr_{\mu_{V j \to i(j)}}[x_j = s_j]\, \psi_{i(j) j}(\sigma_{i(j)}, s_j)$$

$$= \psi_{i_1 i_2}(\sigma_{i_1}, \sigma_{i_2}) \prod_{j \in \partial(i_1,i_2)} \sum_{s \in \{\pm 1\}} \Pr_{\mu_{V j \to i(j)}}[x_j = s]\, \psi_{i(j) j}(\sigma_{i(j)}, s)$$

$$\propto \psi_{i_1 i_2}(\sigma_{i_1}, \sigma_{i_2}) \prod_{j \in \partial(i_1,i_2)} \overline{m}_{\sigma_{i(j)}}^{j \to i(j)} \propto \psi_{i_1 i_2}(\sigma_{i_1}, \sigma_{i_2}) m_{\sigma_{i_1}}^{i_1 \to i_2} m_{\sigma_{i_2}}^{i_2 \to i_1}$$

Belief Propagation

Theorem (BP is exact on trees**).** If G is a tree, belief propagation exactly recovers the marginals of the Gibbs measure

- If G is a general graph, BP is very hard to analyze rigorously
- BP can work when the graph "locally" looks like a tree
- Suppose G is a random graph where every edge is included with probability c/n for $c = O(1)$
- For any vertex v, it has $\approx c$ descendants; and $\approx c^d$ descendants of distance $\leq d$
- Since each descendant is a uniformly random vertex, the probability that there is no path from v to itself of length $\leq d$ is

$$\left(1 - \frac{1}{n}\right)^{c^d} \approx e^{-c^d/n} \to 1 \quad \text{when} \quad c^d \ll n, \qquad \text{i.e.,} \qquad d \ll \log n$$